Joshua N. **Grant**

Geospatial Systems Architect

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Objective

Transform theories, ideas, and research into production-scale geospatial, cloud, and data solutions. Passionate about scalable architectures, machine learning, and automation.

Skills

Languages: Python, R, JavaScript, SQL, Bash, LaTeX
Cloud: AWS, GCP, Azure
Data Eng.: Airflow, Kafka, Prefect, Hadoop
Databases: PostgreSQL, MongoDB, MySQL, Snowflake
GIS: ArcGIS, GeoPandas, PostGIS, GDAL, QGIS
DevOps: Docker, Kubernetes, Ansible, GitHub Actions
Visualization: Shiny, Leaflet, Kibana, Plotly
ML/AI: TensorFlow, Scikit-learn, NLP, AI Deployment
Version Control: Git, GitHub, GitLab

Experience

Geospatial Systems Architect *Oak Ridge National Lab*
2023–Present

- Built cloud-native pipelines for geospatial IoT data processing (Airflow, AWS, PostGIS).
- Designed a geoparquet-based data warehouse for scalable risk analysis.
- Developed geospatial ML models for real-time parcel monitoring.

Senior Software Engineer – Data Engineering *Bold Penguin (Remote)*
2022–2023

- Migrated deployments from Prefect 1 to Prefect 2, automating data workflows.
- Developed scalable ETL pipelines integrating AWS Lambda Fargate.
- Designed event-driven microservices for document processing.

Data Engineer *Oak Ridge National Lab*
2019–2022

- Led geospatial ETL pipeline development for large-scale raster datasets.
- Built machine learning models to detect social media misinformation.

Projects

Geospatial Decision Support Tool
Developed a real-time geospatial analytics platform for risk monitoring. **Technologies:** Python, FastAPI, Kafka, PostGIS, Leaflet, Docker, AWS.

Automated Data Ingestion Pipeline
Designed deployed a cloud-based ETL pipeline for large-scale raster data. **Technologies:** Airflow, Prefect, PostgreSQL, AWS S3, GDAL, Kubernetes.

Education

M.S. Plant Sciences – Molecular Genetics
University of Tennessee, 2017, GPA: 3.72/4.0 **B.S. Plant Sciences – Biotechnology**
University of Tennessee, 2014, Magna Cum Laude, GPA: 3.74/4.0

References

Available upon request.